

Predicting Student Depression using Machine Learning

Authors:

Thabit Bustanji (202211235)

Ayman Al Halabi (202210348)

Jawad Al Haj (202210780)

Introduction

Depressive disorder (also known as depression) is a common mental disorder in which people feel a persistence in low mood or a loss of pleasure or interest in activities that were previously enjoyable for extended periods of time.

Around 3.8% of the population suffers from depression including 5% of adults, (4% of men, and 6% of women).

Depression is 50% more common among women than among men.

Similar Works

Detecting depression using an ensemble classifier based on Quality of Life scales by X. Tao, O. Chi, P. J. Delaney, L. Li, and J. Huang.

Methodology

1-Dataset: The dataset was acquired using Kaggle, an online platform for data science and machine learning. The dataset provides synthetically generated data that simulates the real world. It is also an imbalanced dataset. It includes features such as demographic information and academic information.

Methodology

2-Data Preprocessing: This section begins with data cleaning. The dataset was relatively clean, however some cleaning was necessary to handle outliers and null values. For example, the "fillna()" function was used to fill null values in the dataset, with either the mode for categorical values or the median for continuous values. The "loc()" and "drop()" functions were used to handle out of range data, or non-meaningful data.

Methodology

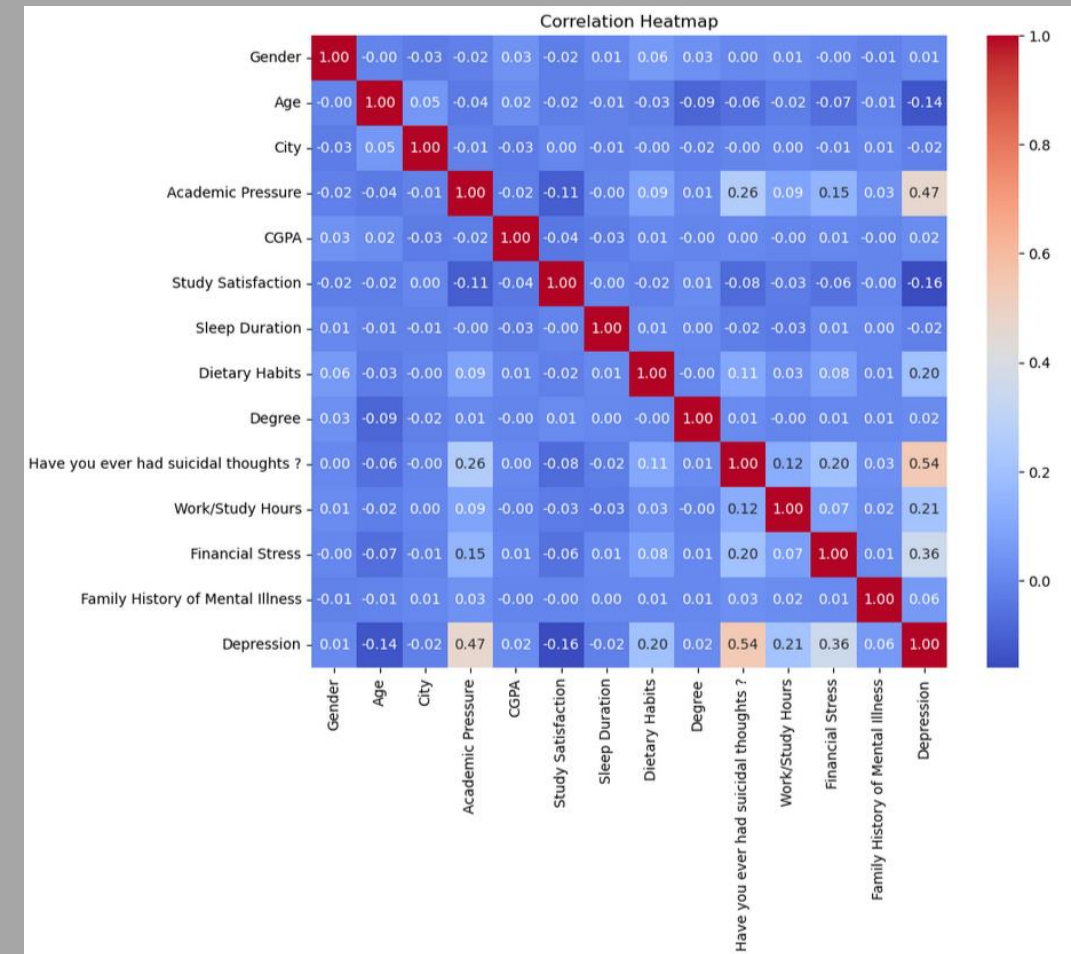
2-Data Preprocessing: Label encoding was used to transform categorical values into numerical values, without altering the inherent weight of the original values.

Methodology

2-Exploratory Data Analysis:

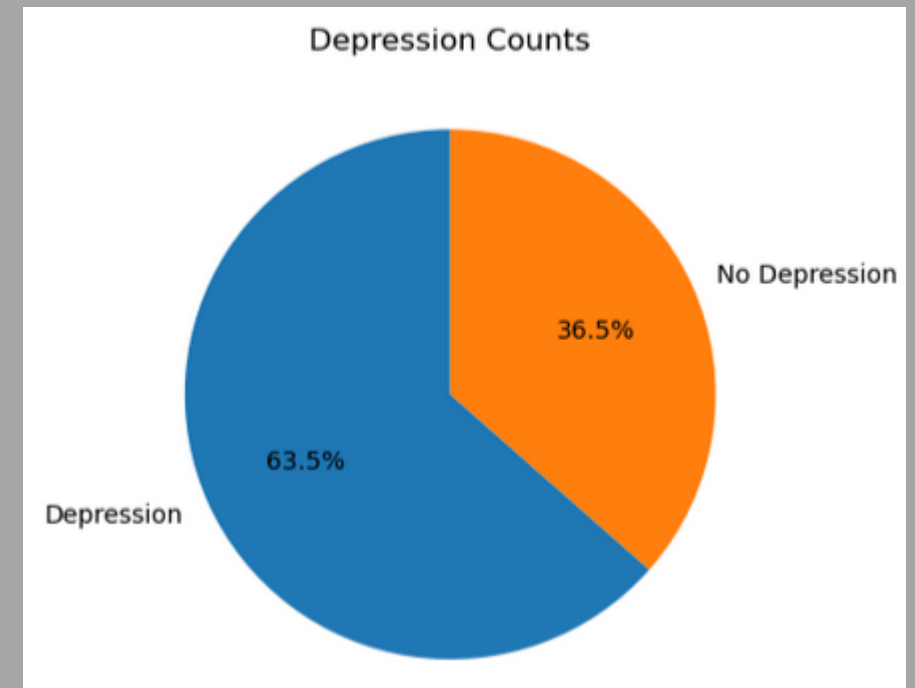
As can be seen in this figure having suicidal thoughts is the most related feature to depression, This is consistent with the fact that suicidal thoughts are a prevalent symptom of depression.

Other interesting correlations can be made, such as higher academic pressure with depression, where students who feel more pressure academically are more likely to feel depressed.



Methodology

This figure shows the imbalance in the dataset.



Methodology

3-Machine Learning: Machine Learning is a way to predict values based on certain features. It can be used to perform complex predictions that would otherwise require a lot of time and effort.

Methodology

Random Forest: Random Forest is a widely used ensemble learning technique for classification and regression tasks. It creates multiple decision trees during training and combines their outputs using majority vote for classification, and average of the trees for regression. Random Forest improves accuracy and helps prevent overfitting.

Methodology

Support Vector Machine: Support vector machine is a supervised machine learning model designed for classification and regression tasks. It works by defining the hyperplane that separates data points of different classes, in the case of classification, the optimal hyperplane is determined by maximizing the margin to reduce overfitting, without causing classification errors or maybe cause it, in the case of non perfectly separable data, the support vector machine is mathematically complex and is a computationally expensive model, designed for classification and regression tasks.

Methodology

Boosting: Boosting is one of the most popular machine learning techniques and can be considered a way of enhancing predictive models' accuracy by combining several weak learners' predictions. Models here are trained sequentially, each trying to correct the errors the previous one makes, hence reducing bias. This is the reason why boosting has gained special recognition and may bring huge performance improvement for the models that are rather weak.

Methodology

Bagging: Bagging is an ensemble learning technique that reduces the variance of an estimate by bootstrapping the data set and separating it into several parts and combining its results after generating a prediction model. The model is built from multiple weaker models and aims towards higher accuracy.

Methodology

Oversampling: Oversampling was used in the dataset in order to balance it. Oversampling can be helpful to prevent models from becoming biased to the majority class, preventing them from accurately classifying the minority class. Random oversampling was used using the imblearn library. Random oversampling duplicates random minority class instances to balance the dataset. This proved to be useful as will be discussed in the results section.

Results

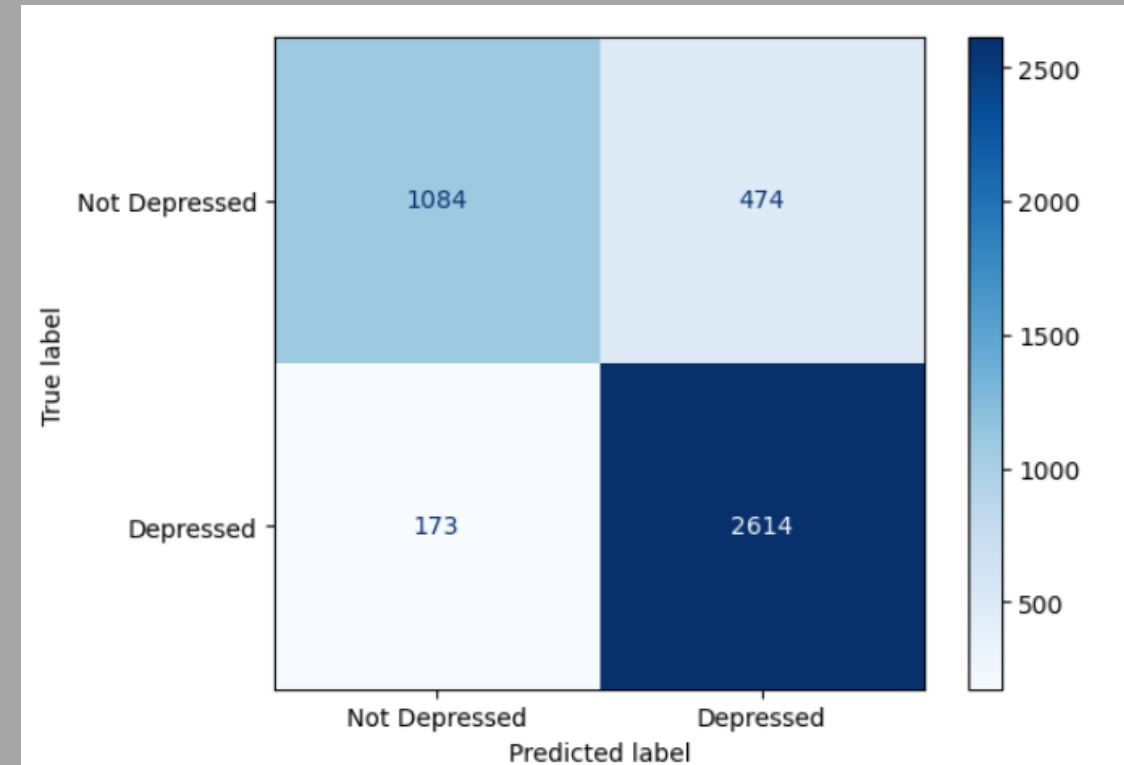
1-Before Oversampling: Before oversampling, most of the models results were close to each other, with some slight differences, however, xgboost will be considered as the best performing model due to its slightly better precision. Table shows the results of the most relevant models.

TABLE I
MODEL RESULTS BEFORE OVERSAMPLING

Model	Accuracy	Recall	Precision	F1-Score
Random Forest	0.84	0.84	0.84	0.84
XGBoost	0.85	0.85	0.85	0.84
SVC	0.85	0.85	0.84	0.84
Extra Trees	0.83	0.83	0.83	0.83

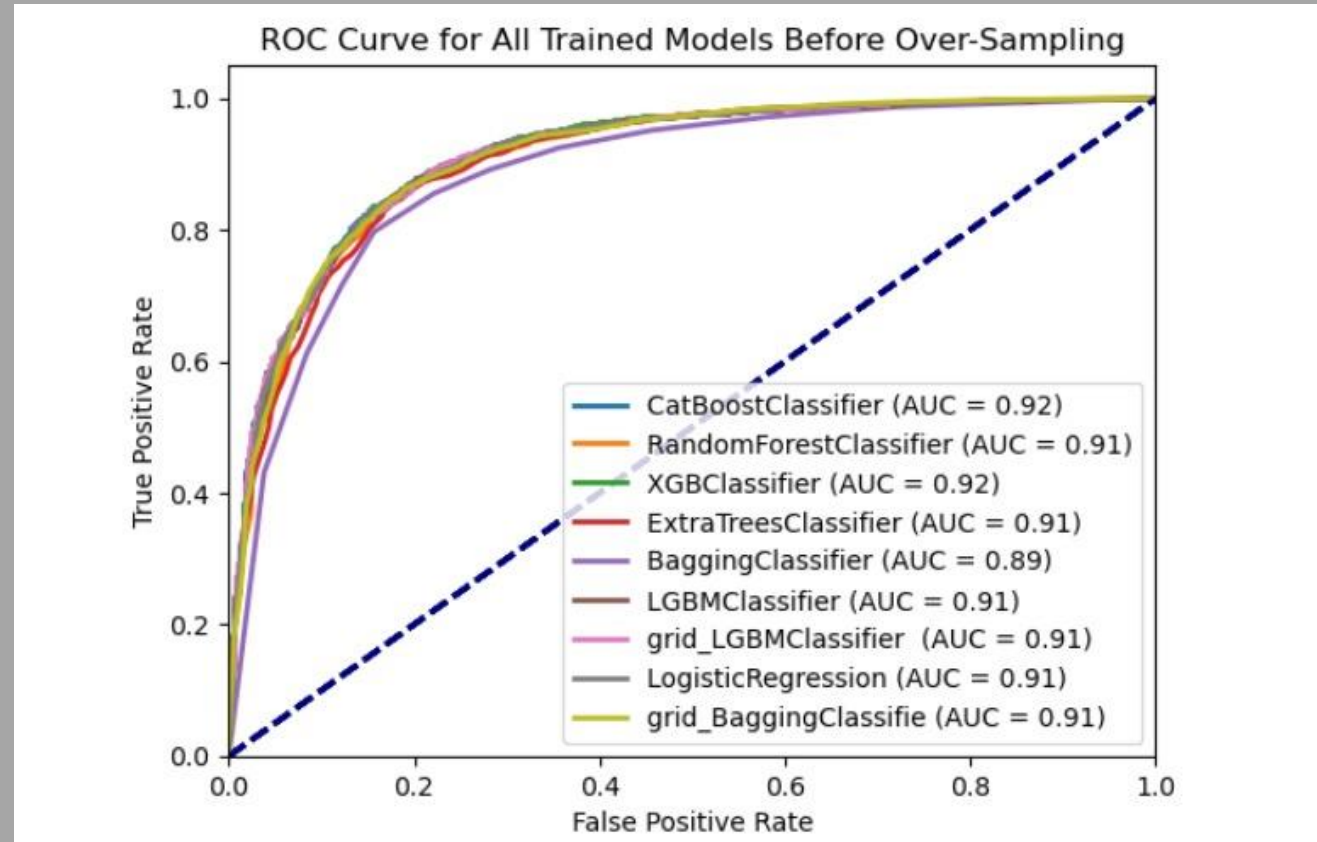
Results

1-Before Oversampling: As can be seen in the figure, the model was able to correctly identify most of the cases, however, bias to the majority class can be seen, as the model seems to be detecting the students with depression better than the students without depression.



Results

1-Before Oversampling: The ROC curve is a graphical representation used to evaluate the performance of a binary classifier. The higher the AUC, the better the model is. As can be seen in figure 12, XGBoost achieved the highest AUC. This aligns with the previously discussed results.



Results

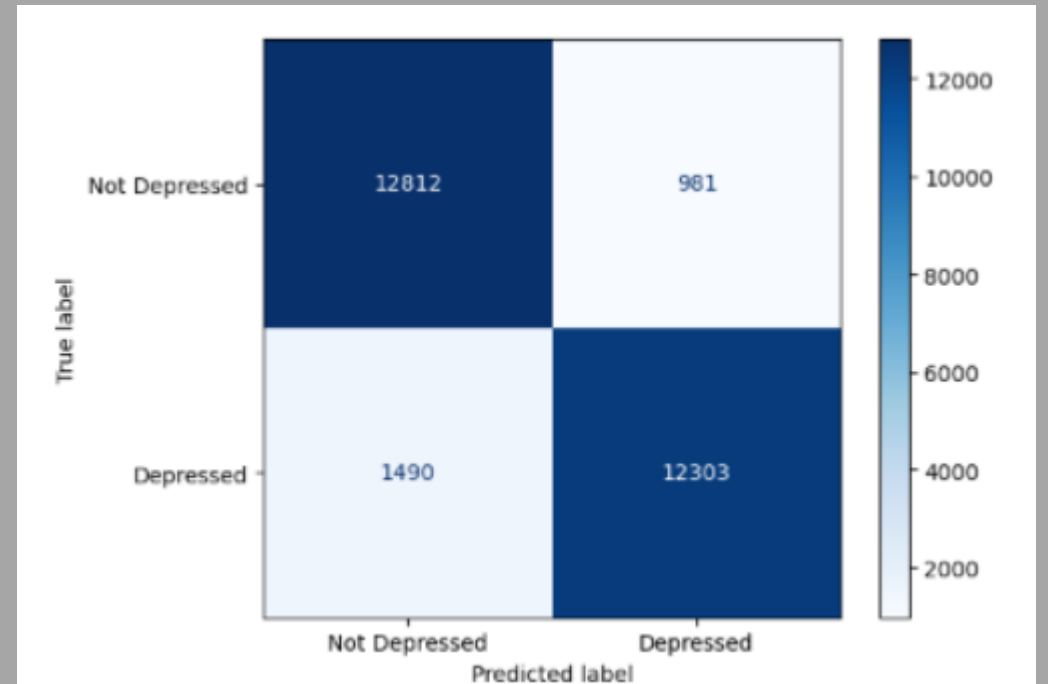
2-After Oversampling: After oversampling, random forest and extra trees obtained excellent results compared to the rest of the classifiers, however, Extra Trees performed the best in all metrics, Table II shows the results of the most relevant models.

TABLE II
MODEL RESULTS AFTER OVERSAMPLING

Model	Accuracy	Recall	Precision	F1-Score
CatBoost	0.84	0.84	0.84	0.84
Random Forest	0.90	0.90	0.90	0.90
XGBoost	0.84	0.84	0.84	0.84
SVC	0.83	0.83	0.83	0.83
Extra Trees	0.91	0.91	0.91	0.91

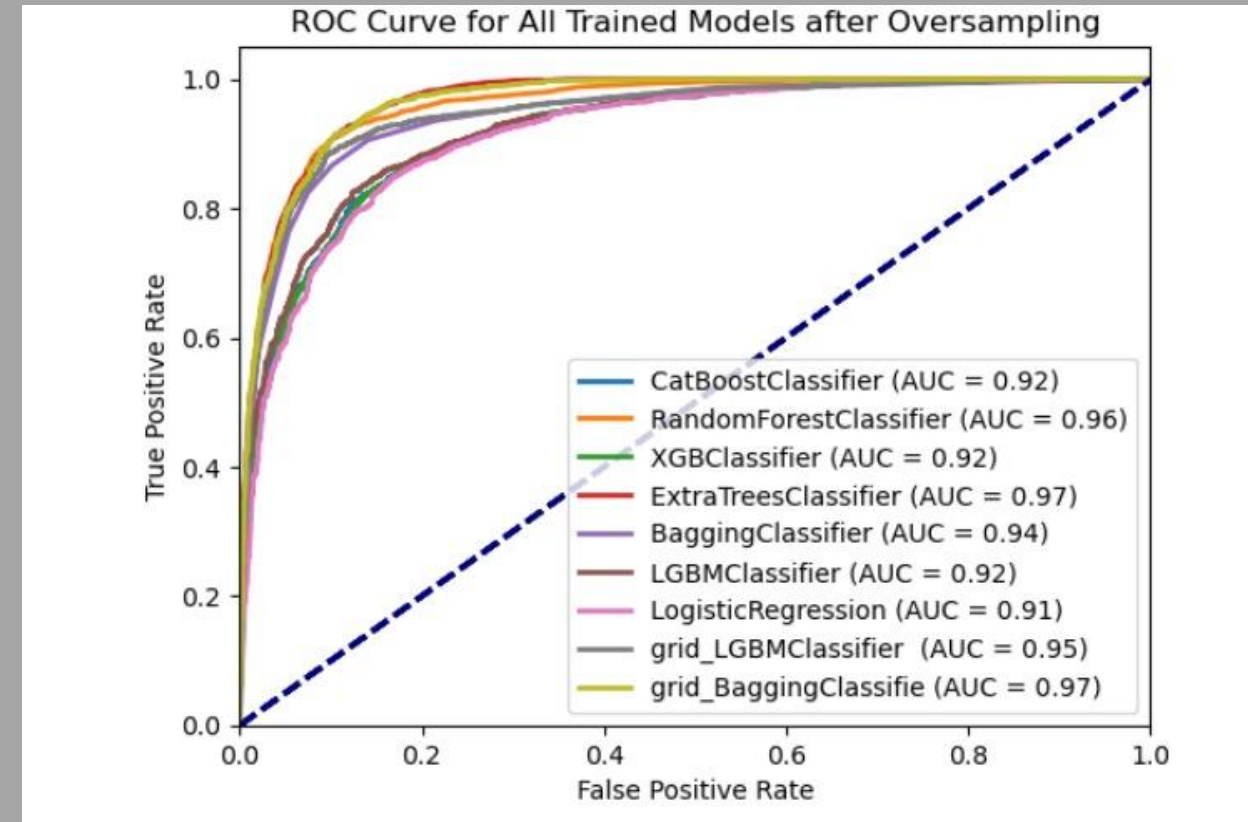
Results

2-After Oversampling: Extra Trees did not achieve a significant result before oversampling, however, after oversampling it was able to achieve the best result. The figure shows the confusion matrix for extra trees classifier, as it was the best performing model.



Results

2-After Oversampling: The figure represents the ROC curve after oversampling. As can be seen, most of the models have shown improvements, particularly the extra trees, bagging and random forest classifiers. This is consistent with the previous results and proves the importance of oversampling overall, and especially in bagging style models.



Conclusion

The paper evaluated multiple ML models using supervised machine learning, to predict student depression based on multiple factors academic and non-academic. After oversampling, extra trees achieved the best results with 91% accuracy and 91% f1-score.

Significant improvements have been made in this field, however more can still be done. Not enough information about depression is gathered from students, which can limit the creation of accurate models about this topic. Moving forward, research could improve analysis of depression, gather more information from students and perform better statistical analysis to improve detection of depression.